THE CLOSURE OF THE EXTENSION OF AN IDEAL IS THE KERNEL OF THE MAP TO A COMPLETED TENSOR PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one chain of identifications of limits; it lies inside the counted span and is reproduced verbatim.

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Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The closure of an ideal is the intersection of its sums with a fundamental system of open ideals. Same chapter, Lemma lemma-closed, invoked at proof lines 1174-1176. Gives the first description of the closure.
- The completed tensor product as a limit of tensor products of quotients. Same chapter, the description of the completed tensor product used at proof lines 1176-1181. Gives the second description, which is then rewritten as a limit of quotients of the target.
- Continuity of the map gives containment of a member of one fundamental system in a member of the other. Same chapter, Example example-what-does-it-mean, invoked at proof lines 1181-1183. Simplifies the limit so that the two descriptions agree.

Source reference. Original definition 0AMR: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 887–887. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AMS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 922–922. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $A \to B$ be a continuous map of linearly topologized rings. Let $I \subset A$ be an ideal. The closure of IB is the kernel of $B \to B\widehat{\otimes}_A A/I$.*

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Proof. Let J_μ be a fundamental system of open ideals of B. The closure of IB is $\bigcap(IB + J_\lambda)$ by Lemma 0AMS. Let I_μ be a fundamental system of open ideals in A. Then

$$B\widehat{\otimes}_A A/I = \lim(B/J_\lambda \otimes_A A/(I_\mu + I)) = \lim B/(J_\lambda + I_\mu B + IB)$$

Since $A \to B$ is continuous, for every λ there is a μ such that $I_\mu B \subset J_\lambda$, see discussion in Example 0AMR. Hence the limit can be written as $\lim B/(J_\lambda + IB)$ and the result is clear. $\square$